



INTERNATIONAL EXAMINATION CENTER

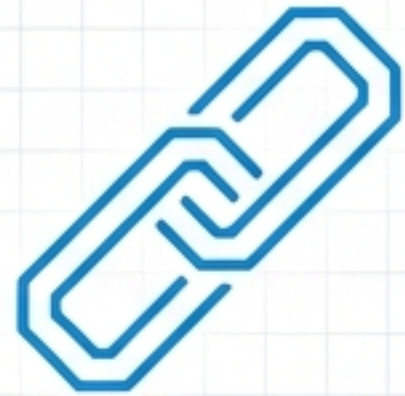
HEADQUARTERS & BRANCH ENTERPRISE NETWORK ARCHITECTURE

ENTERPRISE NETWORKING, SECURITY & AUTOMATION

PROJECT TEAM: SALAHALDEEN THABIT, YAQOOB AL-MULAIKY,
ABDULKAREEM MOTHANA, AHMED AL-WHASHI

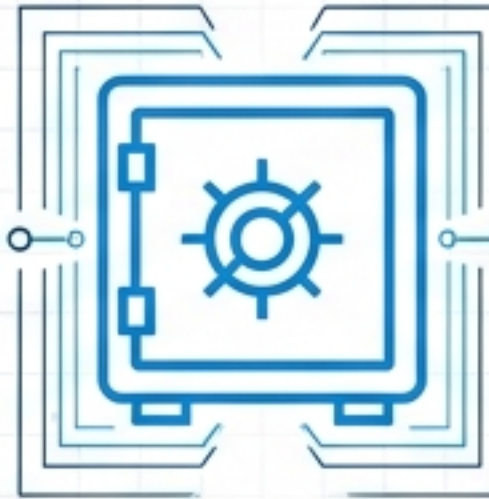
INSTRUCTOR / EVALUATOR: DR. AISHA AL-HADAM

Securing the High-Stakes Examination Environment



Absolute Availability

Continuous operations are mandatory. A single network drop voids an exam in progress, causing immense operational and reputational damage.



Zero-Trust Data Security

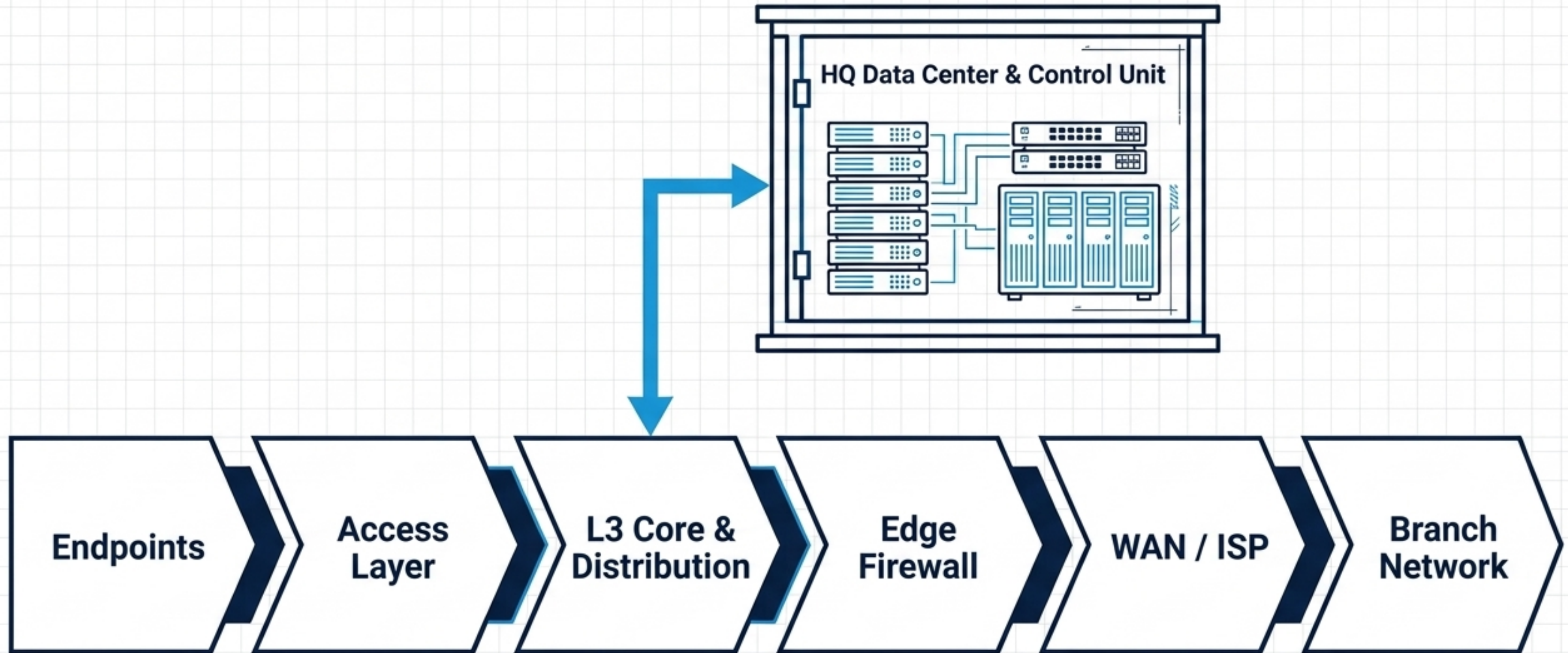
Strict internal boundaries prevent leaks or tampering of proprietary examination content and highly sensitive Personal Identifiable Information (PII).



Centralized Manageability

Operating across Headquarters and Branch locations requires an architecture optimized for centralized monitoring, rapid troubleshooting, and scalable growth.

Enterprise Logical Architecture Blueprint



Physical Design & Infrastructure Layout

Implementation Specifications

HQ Ground Floor

- Dedicated Data Center and Control Unit isolated on the ground floor to ensure strict physical access security.

Distribution & Access

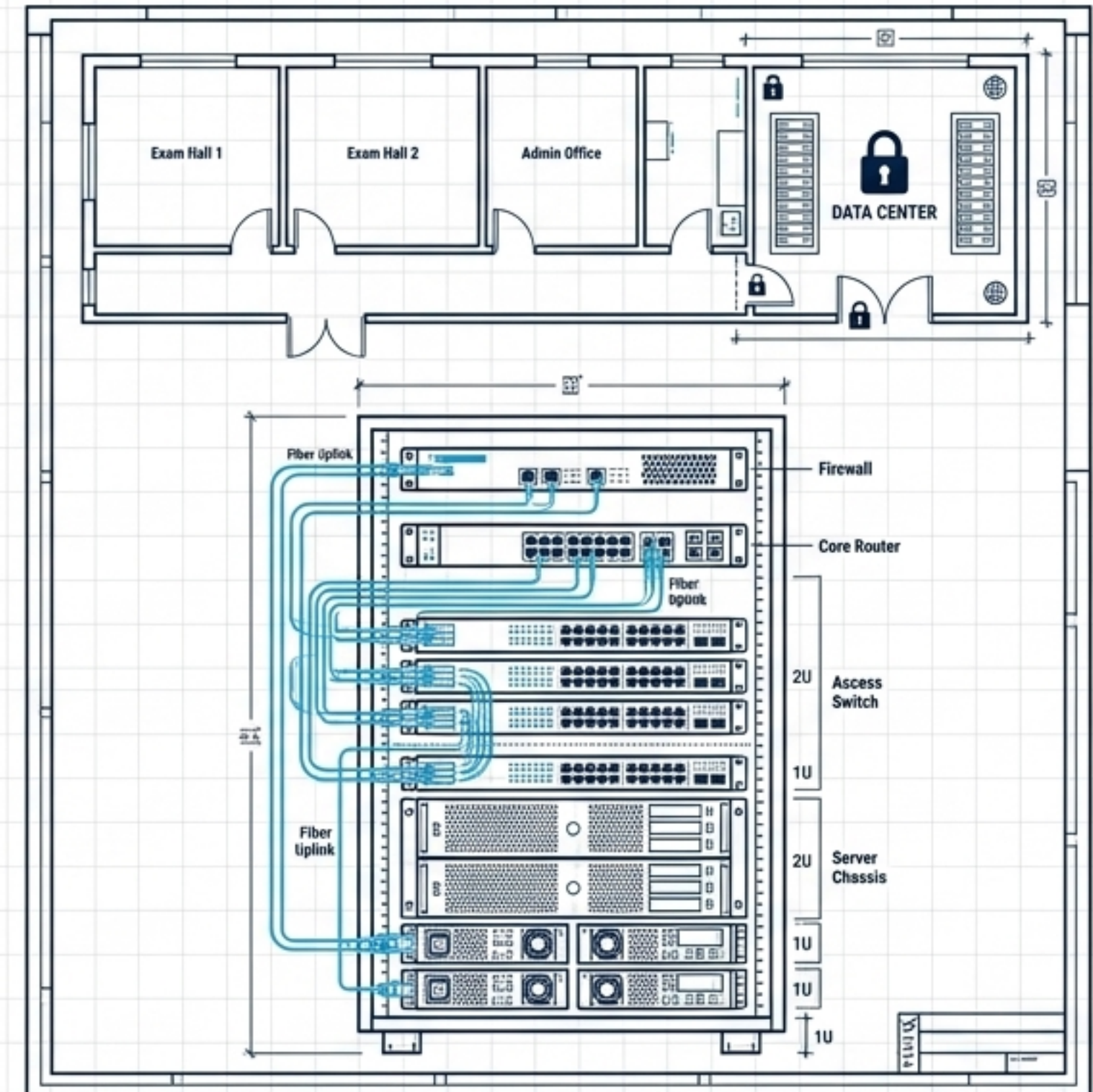
- Floor racks are strategically positioned across the facility to minimize cable runs and optimize cooling.

High-Speed Backbone

- Direct fiber uplinks connecting all Access switches to the Core, guaranteeing high throughput for concurrent exam sessions.

Hardware Organization

- Structured rack placement for Servers, Firewalls, and Routers to optimize cable management and power redundancy.



Strategic Network Segmentation (VLANs)

Corporate Zone	Operations Zone	Infrastructure Zone
 VLAN 10: ADMIN	 VLAN 30: TEST_CENTER	 VLAN 60: SERVERS
 VLAN 20: EMPLOYEES	 VLAN 40: CCTV	 VLAN 70: MANAGEMENT
 VLAN 80: GUEST	 VLAN 50: VOICE	 VLAN 100: DMZ
	 VLAN 90: PRINTERS_IOT	 VLAN 999: NATIVE

Segmentation shrinks broadcast domains, isolates the highly sensitive TEST_CENTER from all general traffic, and secures management protocols from unauthorized lateral movement.

Routing Mechanics & High Availability

Mechanism to Benefit



Layer 3 C9300 Core



Drives **inter-VLAN routing (SVI)** at hardware speeds, offloading processing overhead from the edge router.



OSPFv2 (Area 0)



Provides dynamic, fast-converging routing across the enterprise backbone, adapting automatically to topology changes.



EtherChannel



Aggregates multiple physical links to prevent bandwidth bottlenecks and ensures single-cable failures go unnoticed by users.

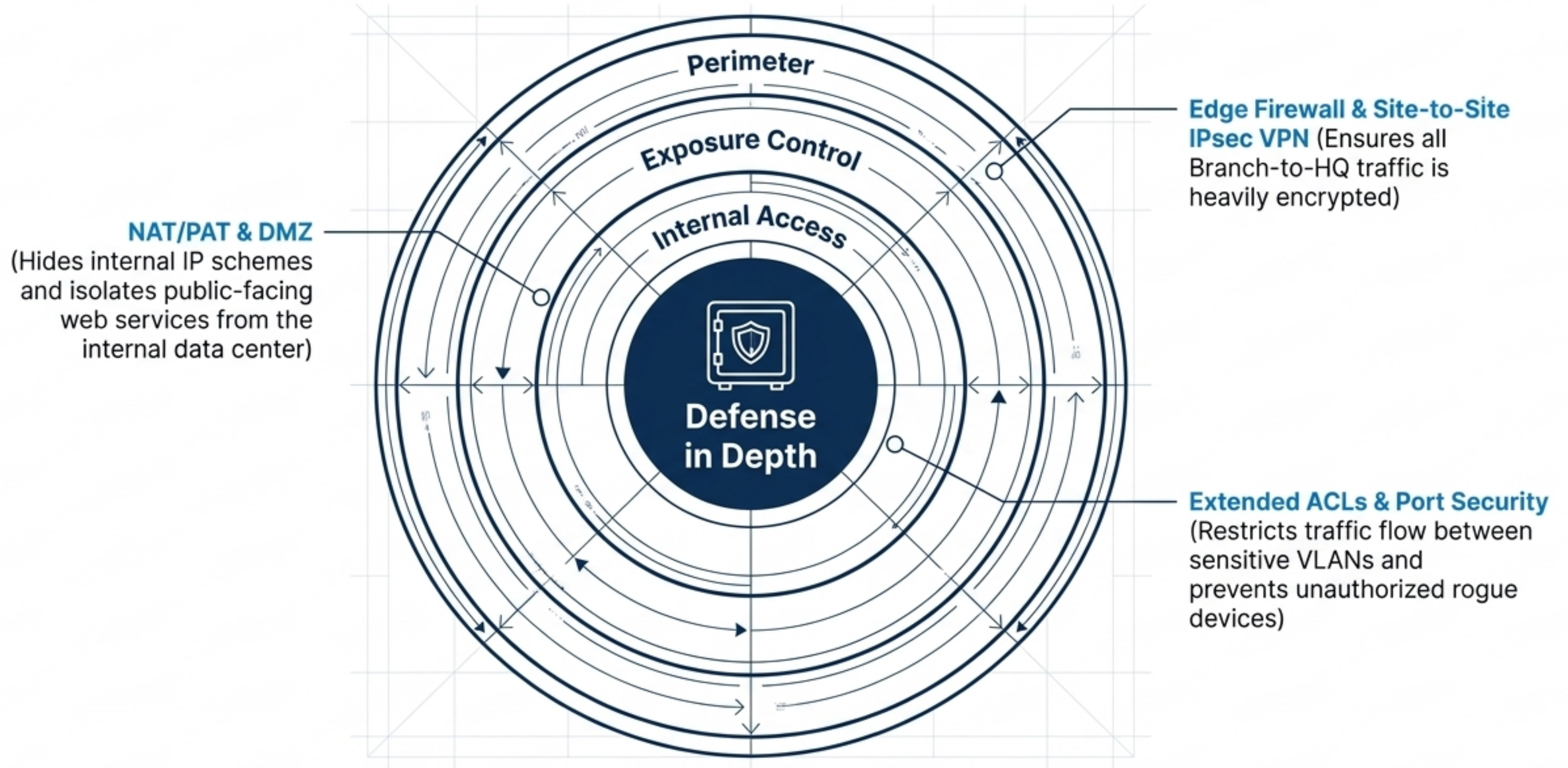


Redundant WAN



Guarantees continuous branch connectivity and exam syncing even in the event of a primary ISP failure.

Layered Security Architecture



Neutralizing threats across multiple, independent layers.

Enterprise Services & Manageability

Identity & IP Infrastructure		
 AD DS Centralized Access Control	 DHCPv4 / DHCPv6 Dynamic IP Allocation	 DNS Internal Name Resolution
Operations & Voice		
 Web Server DMZ Services	 IP-PBX Internal Communications	 CCTV / NVR Physical Facility Monitoring
Manageability & Auditing		
 NTP Crucial time-syncing for accurate security logs	 Syslog, SNMPv3, Zabbix Secure, real-time monitoring	 Centralized Backup Disaster Recovery

Verification, Status & Investment

- > **show ip ospf neighbor**
(Validates core routing adjacencies)
- > **show ip route ospf**
(Confirms dynamic pathing)
- > **show etherchannel summary**
(Verifies link aggregation states)
- > **show ip nat translations**
(Confirms exposure control)
- > **show ntp status**
- > **show logging**
(Verifies audit readiness)

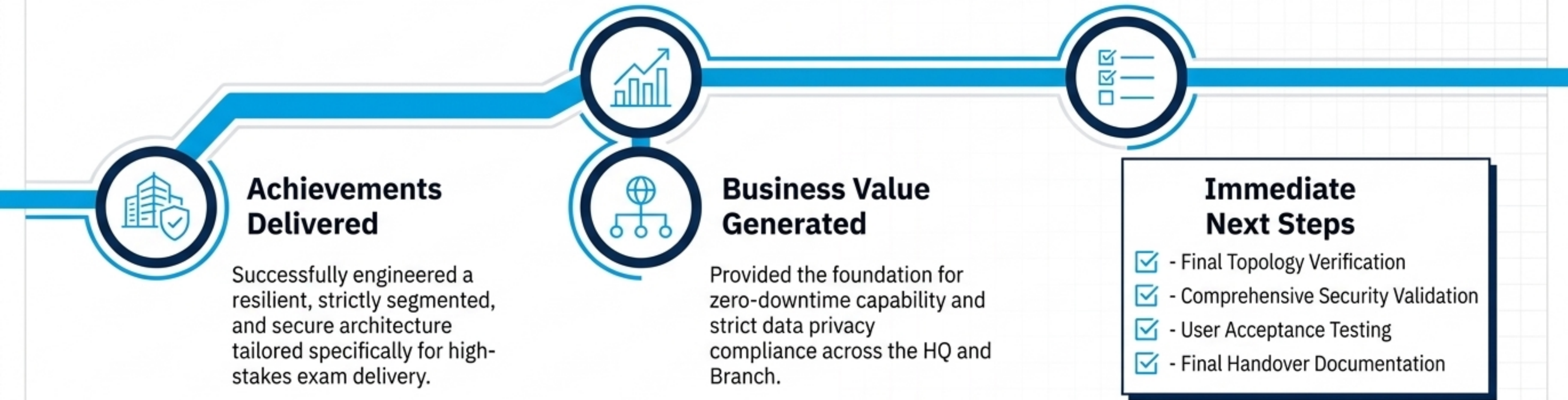
Documented & Verified

- [✓] HQ/Branch Topologies
- [✓] VLAN Matrix
- [✓] Base Security Configurations

Pending / TBD

- [⌚] Final Acceptance Testing
- [⌚] End-to-End Security Penetration Validation
- [⌚] Hardware Investment Total
(Pending Vendor Quotes)

Enterprise Readiness & Next Steps



A comprehensive design successfully simulating and securing a true Enterprise Network environment.